

SMART SEA BELT

Smart Wearable Safety System for Fishermen

PROJECT REPORT

Submitted in partial fulfilment of the requirements for the award of the
degree of

MASTER OF BUSINESS ADMINISTRATION (MBA)

Rohini College of Engineering and Technology
Kanyakumari

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Academic Year 2026–2027

CERTIFICATE

This is to certify that the project report entitled “SMART SEA BELT – Smart Wearable Safety System for Fishermen” is a Bonafide record of the project work carried out by the student team of Rohini College of Engineering and Technology, Kanyakumari, during the academic year 2026–2027, under the guidance and supervision of the concerned faculty mentor.

The project work presented in this report is submitted in partial fulfilment of the academic requirements of the Master of Business Administration programme.

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DECLARATION

We hereby declare that the project report entitled “SMART SEA BELT – Smart Wearable Safety System for Fishermen” is our original work carried out as part of the academic requirements of the Master of Business Administration programme at Rohini College of Engineering and Technology, Kanyakumari.

We further declare that the information, design concepts, analysis, and documentation presented in this report have been prepared for academic and prototype-development purposes. Wherever external concepts or technical references have been used, they are acknowledged in the references section.

Place: Palkulam

Date: 22 - 07 - 2026

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ACKNOWLEDGEMENT

We express our sincere gratitude to the management, Principal, Head of the Department and faculty members of Rohini College of Engineering and Technology, Kanyakumari, for providing us with the opportunity and facilities to undertake this project.

We are especially grateful to our project mentor for continuous guidance, constructive suggestions, and encouragement throughout the development of the Smart Sea Belt concept. The mentor's support helped us refine the problem statement, system architecture, prototype approach, and presentation of the innovation.

We also thank our team members, peers, and all individuals who contributed ideas, technical feedback, and encouragement during the project development process.

Finally, we thank our parents and well-wishers for their constant support and motivation.

ABSTRACT

The Smart Sea Belt is a proposed intelligent marine safety system designed to improve the safety of fishermen and other small-vessel occupants during emergency situations at sea. Marine environments can expose individuals to risks such as accidental falls overboard, loss of balance, delayed emergency communication, sudden health-related incidents and difficulty in reaching a safe location after an emergency.

The proposed system integrates wearable sensing, wireless communication, emergency flotation and an automatic rudder diversion mechanism into a coordinated safety architecture. Each Smart Watch unit can monitor parameters such as heart rate and motion and provides an SOS interface. Bluetooth Low Energy (BLE) is used for communication between wearable units and a centralized Boat Hub. The Boat Hub combines the wearable data with GPS information and forwards relevant emergency information through LoRa communication to a Base Unit. The Base Unit supports monitoring and emergency alerts.

For an emergency involving a person overboard or a critical safety event, the system is designed to support two physical safety responses: an automatically deployable flotation mechanism and an automatic rudder diversion mechanism. The flotation subsystem is intended to assist the person in maintaining buoyancy, while the rudder mechanism is intended to redirect the boat toward the relevant emergency location or safer recovery direction, subject to the vessel's operating conditions and control logic.

The proposed architecture follows the sequence “Sense → Process → Detect → Communicate → Alert”. The system also provides a foundation for an IoT-enabled mobile application through which boat operators, authorized monitoring personnel and rescue stakeholders can view alerts and status information. The estimated prototype development cost is ₹58,000, covering the base unit, Boat Hub, five wearable units, integration/testing and marine protection.

This project report presents the problem background, objectives, literature-oriented context, proposed architecture, components, working methodology, automatic rudder mechanism, software and communication design, cost structure, feasibility, limitations, future scope and commercialization considerations.

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1. INTRODUCTION

Fishing and small-vessel operations are important for economic and social activities for coastal communities. People working at sea may encounter rapidly changing environmental conditions and emergency situations where immediate assistance is difficult. A safety system for such environments should therefore do more than provide a conventional wearable alarm; it should support detection, communication, physical protection and coordinated response.

The Smart Sea Belt is conceptualized as a connected marine safety ecosystem. The wearable component is intended to remain with the individual, while the Boat Hub acts as the central processing and communication point within the vessel. A Base Unit provides an additional monitoring endpoint. The design combines local sensing with longer-range communication and physical emergency-response mechanisms.

The project focuses on a practical chain of response: detect an abnormal event, identify the affected wearer, communicate the event, activate the appropriate safety mechanism and provide information for recovery. The concept is designed as a prototype and should undergo marine-environment testing, control-system validation and safety certification before operational deployment.

1.1 Background

Conventional safety equipment such as life jackets, whistles, emergency lights and manual communication devices can provide important protection, but their effectiveness may depend on timely human action. An intelligent system can complement such equipment by automating selected detection and alert functions.

1.2 Project Concept

The Smart Sea Belt consists of wearable Smart Watch units, a Boat Hub, a Base Unit, communication links, emergency flotation and an automatic rudder diversion assembly. The core principle is to reduce the time between emergency detection and initiation of a response.

1.3 Scope

- Design of an intelligent wearable marine safety architecture.
- Integration of heart rate, motion, and SOS sensing.
- BLE communication from wearable devices to the Boat Hub.
- GPS-supported location awareness at the Boat Hub.
- LoRa communication between Boat Hub and Base Unit.
- Emergency flotation activation through a controlled release mechanism.
- Automatic rudder diversion concept for emergency response.
- IoT/mobile monitoring concept for authorized users.
- Prototype cost estimation and validation planning.

2. PROBLEM STATEMENT

A person working on a fishing boat may face an emergency in which conventional manual response is delayed. A fall, abnormal motion pattern, sudden health-related event or direct SOS request may require rapid communication to the boat operator or monitoring personnel. If the person is separated from the vessel, maintaining buoyancy and communicating the approximate location becomes critical.

The project therefore addresses the need for an integrated safety solution that can sense selected emergency indicators, communicate them to a central unit, support flotation and assist the vessel in changing direction toward a recovery path.

2.1 Key Problems Identified

- Delayed awareness of an individual emergency.
- Difficulty in identifying which wearable user requires assistance.
- Limited communication range of short-range wearable technologies.
- Dependence on manual emergency activation and response.
- Risk of losing the individual's position during a person-overboard event.
- Need for an integrated safety response rather than a single-purpose wearable.
- Challenges of operating electronic systems in a marine environment.

3. OBJECTIVES

No.	Objective
1	Develop a wearable-based marine safety concept for fishermen.
2	Monitor heart rate and motion-related safety indicators.
3	Provide a direct SOS interface for emergency activation.
4	Use BLE to connect Smart Watch units with the Boat Hub.
5	Use GPS at the Boat Hub for location awareness.
6	Use LoRa for long-range communication to the Base Unit.
7	Provide a controlled emergency flotation mechanism.
8	Develop an automatic rudder diversion concept for emergency response.
9	Create an IoT/mobile monitoring framework for alerts.
10	Evaluate prototype cost, feasibility, risks and future scalability.

4. NEED AND SIGNIFICANCE

The significance of Smart Sea Belt lies in combining multiple layers of safety into one coordinated architecture. Instead of treating wearable sensing, communication, flotation and boat control as independent functions, the project proposes a connected response pathway.

4.1 Beneficiaries

- Fishermen and crew members operating small vessels.
- Boat owners and operators responsible for crew safety.
- Authorized family or monitoring stakeholders through alert systems.
- Rescue and emergency-response personnel, subject to suitable integration.
- Institutions working on marine safety and IoT innovation.

5. EXISTING SYSTEM

Existing marine safety practices commonly rely on personal flotation devices, manual SOS communication, mobile or radio communication, visual observation and GPS-enabled equipment. Each method addresses a particular part of the safety problem. However, the response may still depend on the person or operator recognizing the emergency and taking appropriate action.

Aspect	Conventional Approach	Smart Sea Belt Concept
Detection	Manual observation or manual SOS	Motion/heart-rate indicators + SOS
Identification	Visual/manual	Assigned wearable identity
Short-range link	Usually independent	BLE to Boat Hub
Location	Separate GPS device/system	GPS integrated at Boat Hub
Long-range alert	Radio/mobile/manual	LoRa to Base Unit
Flotation	Manual/standard safety equipment	Controlled deployment concept
Boat response	Manual steering	Automatic rudder diversion concept
Monitoring	Limited/manual	IoT/mobile dashboard concept

6. PROPOSED SYSTEM

The proposed Smart Sea Belt system is organized into four major layers: sensing, processing, communication, and response. The Smart Watch acts as the sensing and user interface layer. The Boat Hub receives data and manages location and emergency processing. The Base Unit provides a longer-range monitoring endpoint. The response layer includes flotation and automatic rudder diversion.

6.1 System Modules

- Smart Watch Module
- Boat Hub Module
- Base Unit Module

- BLE Communication Module
- LoRa Communication Module
- GPS Module
- Emergency Flotation Module
- Automatic Rudder Diversion Module
- IoT/Mobile Monitoring Module

7. SYSTEM ARCHITECTURE

The architecture follows the data and response pathway: Smart Watch → BLE → Boat Hub → LoRa → Base Unit → Monitoring & Emergency Alerts. The Boat Hub also interfaces with the automatic rudder and flotation subsystems according to predefined emergency rules.

Layer	Main Function	Representative Elements
Sense	Collect safety indicators	MAX30102, MPU6050, SOS button
Process	Interpret sensor events	ESP32-C3 / embedded logic
Locate	Provide location context	GPS at Boat Hub
Communicate	Transfer emergency data	BLE + LoRa
Respond	Provide physical safety actions	Flotation + rudder mechanism
Monitor	Present alerts/status	IoT/mobile application

7.1 Data Flow

A Smart Watch periodically measures or detects relevant parameters. The data is transmitted to the Boat Hub through BLE. The Boat Hub associates the data with the corresponding wearer, evaluates emergency conditions, and attaches GPS context. If an emergency rule is triggered, the Boat Hub can initiate local response actions and send an alert through LoRa to the Base Unit.

7.2 Design Principle

The overall design principle is expressed as: Sense → Process → Detect → Communicate → Alert → Respond.

8. HARDWARE COMPONENTS

Component	Purpose	Integration
ESP32-C3	Embedded processing and BLE connectivity	Smart Watch / control node
MAX30102	Heart rate and optical sensing	Smart Watch
MPU6050	Motion and orientation sensing	Smart Watch

SOS Button	Manual emergency request	Smart Watch
GPS Module	Position information	Boat Hub
LoRa Module	Long-range data communication	Boat Hub/Base Unit
Servo Motor	Mechanical actuation	Rudder/flotation control
CO ₂ Valve/Release	Flotation deployment control	Flotation subsystem
CO ₂ Cartridge	Inflation source	Flotation subsystem
Rechargeable Battery	Power supply	Wearable/control units
Marine Protective Enclosure	Environmental protection	System housing

8.1 ESP32-C3

The ESP32-C3 is proposed as a compact microcontroller platform with wireless capability. In the Smart Watch, it can acquire sensor readings, manage the SOS interface, process event conditions and communicate with the Boat Hub using BLE.

8.2 MAX30102

The MAX30102 is proposed for optical heart-rate sensing. The project treats heart-rate information as one supporting indicator rather than a standalone medical diagnosis. Thresholds should be experimentally validated for the intended users and operating conditions.

8.3 MPU6050

The MPU6050 provides accelerometer and gyroscope measurements. These can be used to detect sudden motion, orientation changes or possible fall-like events. The prototype should use multiple readings and timing logic to reduce false alarms.

8.4 Marine Protection

The electronics enclosure should be designed for marine exposure, with suitable sealing, corrosion-resistant materials and a waterproofing strategy. Thermoplastic polyurethane may be considered for flexible protective components, subject to mechanical and environmental testing.

9. SOFTWARE AND COMMUNICATION TECHNOLOGIES

9.1 Bluetooth Low Energy

BLE provides a low-power short-range connection between Smart Watch units and the Boat Hub. Each wearable can have a unique identifier so the Boat Hub can distinguish among multiple crew members.

9.2 LoRa

LoRa is proposed for long-range, low-data-rate communication between the Boat Hub and Base Unit. Emergency messages can contain a compact payload such as device ID, event type, timestamp, battery status and location information where available.

9.3 GPS

GPS is used at the Boat Hub to provide location context. The location information can be attached to emergency alerts and displayed through the monitoring application.

9.4 IoT and Mobile Application

An IoT layer can store and display emergency events, system status and historical information. A mobile application can provide authorized users with a simple dashboard showing active alerts, wearer status and location information.

10. WORKING PRINCIPLE

During normal operation, the Smart Watch remains connected to the Boat Hub through BLE. Sensor readings are periodically acquired. The Boat Hub maintains a list of active wearable IDs and their communication status.

When a wearer presses the SOS button, the Smart Watch generates an emergency event. A motion-based event can also be generated when sensor readings satisfy a predefined detection rule. Heart-rate information may be used as a supporting signal for an abnormal-condition rule. The Boat Hub validates the received event, obtains current GPS information and activates the configured emergency response.

The emergency response can include an alert to the Base Unit, local audible/visual indication, controlled flotation activation and the automatic rudder diversion sequence where the emergency condition and system design permit it.

10.1 Emergency Event Types

Event	Possible Trigger	Response
Manual SOS	SOS button pressed	Immediate alert + configured safety response
Fall-like event	Sudden motion/orientation pattern	Warning/confirmation logic + alert
Communication loss	Wearable disconnected	Status warning / operator attention
Abnormal vital indicator	Configured sensor condition	Warning + confirmation/alert
Person-overboard response	Emergency rule or manual confirmation	Flotation + location + rudder sequence

11. AUTOMATIC RUDDER DIVERSION SYSTEM

The automatic rudder diversion system is a key innovation component of the Smart Sea Belt. Its purpose is to assist the vessel in changing direction during an emergency so that the

vessel can move toward a safer recovery path. The mechanism is conceptual and must be engineered with vessel-specific control limits before real-world deployment.

11.1 Architecture

Stage	Function
Emergency detection	Receive validated emergency event
Location acquisition	Obtain current position and event context
Decision logic	Determine whether rudder response is permitted
Control signal	Send command to actuator/controller
Actuation	Servo or suitable actuator changes rudder position
Stabilization	Monitor vessel direction and actuator state
Recovery mode	Return to operator control under defined conditions

11.2 Rudder Control Logic

The rudder subsystem should not blindly turn the vessel whenever a sensor threshold is crossed. A safer prototype architecture uses validation, operator override, predefined rudder limits and a timeout. The control logic should also include a fail-safe state in which manual steering remains available.

- Validate the emergency event before automatic steering.
- Confirm that the rudder actuator is available and powered.
- Apply a predefined maximum rudder angle.
- Use a controlled movement rather than an abrupt command.
- Provide manual override to the boat operator.
- Stop automatic control after a defined timeout or recovery condition.
- Record the event for later testing and analysis.

12. FLOTATION AND EMERGENCY SAFETY MECHANISM

The Smart Sea Belt concept includes a controlled flotation mechanism intended to improve the wearer's buoyancy during an emergency. A servo-controlled CO₂ release mechanism can be used to activate an inflation system. The mechanism should be physically separated from sensitive electronics and designed to prevent accidental deployment.

The project scope treats the SOS function specifically as an activation input for the flotation mechanism. Location and emergency data transmission are handled through the wearable-to-Boat Hub communication architecture.

12.1 Safety Considerations

- Prevent accidental activation through mechanical and software safeguards.
- Provide a manual override or secondary activation method where appropriate.
- Use marine compatible materials and sealed connectors.
- Test inflation reliability under controlled conditions.
- Validate cartridge compatibility and pressure requirements.
- Conduct wearer safety testing before any field deployment.

13. MOBILE APPLICATION AND IOT MONITORING

The proposed application provides an interface for monitoring emergency conditions. The application can display the list of registered wearable units, current status, emergency events, location information and system alerts.

Screen	Information
Dashboard	Overall vessel and wearable status
Wearer Status	Wearer ID, connection and battery status
Emergency Alert	Event type, timestamp and location
Map	GPS position and recovery context
History	Previous events and response records
Device Settings	Registration and threshold configuration
System Health	Battery, connectivity and actuator status

13.1 AI-Assisted Features

Future versions may use AI-assisted anomaly detection to identify unusual motion or physiological patterns. Such a model should be trained on appropriate data and evaluated for false positives and false negatives. AI output should support, rather than replace, established emergency procedures.

14. METHODOLOGY AND FLOWCHARTS

14.1 Overall Methodology

- Identify marine safety problem.
- Study existing safety practices and technology options.
- Define functional requirements.
- Design wearable sensing architecture.
- Design Boat Hub and Base Unit communication.
- Develop emergency response logic.
- Design flotation and rudder mechanisms.

- Develop prototype hardware and software.
- Conduct controlled testing.
- Evaluate performance, risks and feasibility.
- Refine design and document outcomes.

14.2 Overall Flow

START → Initialize Smart Watch → Read Sensors → Check SOS / Motion / Vital Indicators → Send Event via BLE → Boat Hub Validation → Acquire GPS → Emergency Decision → Send LoRa Alert → Activate Appropriate Safety Response → Monitor → Operator/Rescue Action → END/RESET

14.3 Rudder Flow

Emergency Confirmed → Check Rudder System Health → Check Manual Override → Calculate Safe Rudder Command → Limit Rudder Angle → Actuate Rudder → Monitor Direction/Timeout → Return Control to Operator

15. PROTOTYPE DEVELOPMENT PLAN

Phase	Activities	Expected Output
Phase 1	Requirement analysis	Functional requirements
Phase 2	Circuit and module selection	Component architecture
Phase 3	Wearable prototype	Smart Watch prototype
Phase 4	Boat Hub/Base Unit	Communication prototype
Phase 5	Flotation subsystem	Controlled deployment prototype
Phase 6	Rudder subsystem	Actuator/control prototype
Phase 7	Software integration	Monitoring application
Phase 8	Testing	Performance and risk results
Phase 9	Documentation	Final report and presentation

16. COST STRUCTURE

Item	Estimated Cost (₹)
Base Unit	16,000
Boat Hub	17,000
Five Smart Watch Units	20,000
Integration and Testing	3,000
Marine Protection / Enclosure	2,000
Total	58,000

The above figures represent the estimated prototype budget for the current project concept. Actual procurement costs can change based on component specifications, supplier, enclosure quality, testing requirements and production quantity.

16.1 Cost Interpretation

The largest cost components are the five wearable units and the Boat Hub. At prototype stage, the objective is functional validation rather than mass-production optimization. A future production version could reduce per-unit cost through component consolidation, custom PCB design, optimized enclosure manufacturing, and volume procurement.

17. FEASIBILITY ANALYSIS

17.1 Technical Feasibility

The individual technologies required for the prototype—microcontrollers, BLE, motion sensing, optical sensing, GPS, LoRa and servo actuation—are widely used embedded-system technologies. The main technical challenge is integrating them into a reliable marine-grade system with low power consumption and controlled failure behavior.

17.2 Economic Feasibility

The estimated prototype cost of ₹58,000 provides a basis for proof-of-concept development. Commercial feasibility would require a separate detailed bill of materials, manufacturing cost analysis, certification costs, maintenance assumptions and customer willingness-to-pay research.

17.3 Operational Feasibility

The proposed system is intended to minimize complex user interaction. The wearer primarily uses the wearable normally and can trigger an SOS when required. The boat operator receives centralized information through the Boat Hub and monitoring interface.

17.4 Environmental Feasibility

Marine conditions create challenges including saltwater corrosion, humidity, water ingress, vibration and temperature changes. These factors must be addressed through enclosure design, material selection and environmental testing.

18. INNOVATION AND UNIQUENESS

The project's innovation is the integration of multiple safety functions into a single connected architecture. The Smart Sea Belt is not limited to a wearable alarm. It combines wearable sensing, BLE connectivity, centralized boat-level processing, GPS context, LoRa communication, flotation support and an automatic rudder diversion concept.

- Multi-wearer monitoring through a centralized Boat Hub.
- Short-range BLE combined with long-range LoRa communication.

- Integrated emergency flotation activation concept.
- Automatic rudder diversion as a physical response layer.
- Potential IoT/mobile monitoring.
- Future AI-assisted anomaly detection.
- Modular architecture suitable for future expansion.

19. ADVANTAGES AND APPLICATIONS

19.1 Advantages

- Faster identification of selected emergency conditions.
- Centralized monitoring of multiple wearable users.
- Long-range alert capability through LoRa architecture.
- Location-aware emergency communication.
- Physical safety response through flotation.
- Potential assistance in vessel redirection.
- Expandable IoT and AI architecture.

19.2 Applications

- Fishing boats
- Small commercial marine vessels
- Coastal field operations
- Marine research teams
- Rescue-support operations
- Training and safety demonstrations
- Future integration with institutional marine-safety networks

20. LIMITATIONS AND RISK CONSIDERATIONS

Risk	Potential Impact	Mitigation
False alarm	Unnecessary response	Multi-sensor validation and confirmation logic
Sensor failure	Missed detection	Self-test and periodic diagnostics
BLE disconnection	Loss of wearable data	Connection-status monitoring
LoRa interruption	Delayed remote alert	Local alarm and retry mechanism
GPS inaccuracy	Incorrect location	Quality checks and fallback procedures
Battery depletion	System shutdown	Battery monitoring and charging schedule

Water ingress	Electronics damage	Marine-rated enclosure/testing
Rudder actuator failure	Unsafe/incomplete diversion	Manual override and fail-safe state
Mechanical failure	Reduced safety	Preventive inspection and redundancy where practical
Human factors	Incorrect response	Simple UI and operator training

The project is a prototype concept and should not be treated as a certified marine safety device without appropriate engineering validation, regulatory review, and field testing.

21. TESTING AND VALIDATION PLAN

Test Area	Test Method	Expected Result
Heart-rate sensing	Compare readings with reference device	Acceptable measurement behaviour
Motion detection	Controlled movement/fall-like tests	Event detected within defined limits
SOS	Press SOS repeatedly under controlled conditions	Reliable event generation
BLE	Test multiple wearables at different distances	Stable connection within design range
LoRa	Test communication over representative distances	Emergency packet received
GPS	Compare location with known points	Location within expected accuracy
Flotation	Controlled activation tests	Reliable deployment without accidental activation
Rudder	Bench/controlled actuator tests	Correct direction and limit response
Power	Continuous operating test	Expected operating duration
Marine enclosure	Water/splash/environment tests	No unacceptable water ingress

21.1 Validation Metrics

- Detection success rate
- False alarm rate
- Emergency message delivery rate

- Alert latency
- GPS location error
- Battery operating duration
- Actuator response time
- Flotation deployment reliability
- System recovery after communication loss

22. FUTURE SCOPE

The Smart Sea Belt can be expanded from a prototype into a more comprehensive marine safety platform. Future development can focus on miniaturization, custom PCB design, stronger marine protection, improved anomaly detection, better energy management, and integration with rescue systems.

Stage	Future Development
Short Term	Prototype refinement, enclosure improvement, controlled testing
Medium Term	Custom PCB, improved power management, mobile application
Advanced	AI anomaly detection, improved rudder control and multi-vessel monitoring
Scale-up	Institutional dashboards, fleet management and rescue integration

- Satellite communication for offshore environments.
- Mesh networking among multiple boats.
- Cloud-based fleet safety analytics.
- Predictive maintenance for vessel safety systems.
- Digital incident reporting.
- Integration with authorized emergency-response organizations.
- Solar or energy-harvesting options for auxiliary power.

23. BUSINESS AND COMMERCIALIZATION MODEL

A potential commercialization model can combine hardware sales with software and service support. The primary customers could include fishing cooperatives, individual boat owners, marine institutions and organizations operating small vessels.

Revenue Stream	Description
Hardware	Smart Sea Belt kits, Boat Hub and Base Unit
Installation	System installation and vessel-specific

	integration
Software	Monitoring dashboard/mobile application
Service	Maintenance, inspection and replacement
Institutional Deployment	Fleet-level monitoring and analytics
Customization	Vessel-specific safety and control integration

23.1 Value Proposition

The proposed value proposition is an integrated safety layer that connects the person, vessel and monitoring point. Instead of depending only on manual communication, the system attempts to automate selected parts of emergency detection and response.

24. PATENT-ORIENTED TECHNICAL DESCRIPTION

The project contains technical concepts that may be considered for intellectual property evaluation. A qualified patentability assessment should be conducted by a prior-art search and detailed claim analysis.

24.1 Core Concept

A marine wearable safety system comprising one or more wearable sensor units, a boat-level communication hub, a long-range communication unit, a location module and one or more emergency response mechanisms, wherein an emergency condition detected by a wearable unit is communicated to the boat-level hub and used to initiate an emergency response.

24.2 Automatic Rudder Concept

A controlled rudder diversion assembly configured to receive an authenticated emergency command and modify rudder position within predefined limits, with manual override and fail-safe conditions.

24.3 Flotation Concept

A controlled flotation assembly associated with the wearable safety system and configured to initiate inflation following a validated emergency input, while maintaining separation between flotation activation and location-data communication functions.

24.4 Patent Drawing Set Suggested

- Overall system block diagram
- Smart Watch internal block diagram
- Boat Hub architecture
- LoRa Base Unit architecture
- Flotation deployment assembly
- Automatic rudder diversion assembly
- Integrated emergency-response flowchart

- System communication sequence

25. PROJECT OUTCOMES

The project produces a structured concept for an intelligent marine safety system that connects wearable sensing, boat-level communication and physical emergency response. It provides a basis for prototype development and further technical validation.

- Defined problem and functional requirements.
- Developed integrated system architecture.
- Selected representative electronic components.
- Defined BLE and LoRa communication pathway.
- Designed flotation response concept.
- Designed automatic rudder diversion concept.
- Prepared IoT/mobile monitoring framework.
- Estimated prototype cost at ₹58,000.
- Defined testing, risk mitigation and future development roadmap.

26. CONCLUSION

The Smart Sea Belt presents an integrated approach to marine safety by combining wearable sensing, emergency communication and physical response mechanisms. The proposed architecture uses Smart Watch units for sensing and SOS interaction, BLE for short-range communication, a Boat Hub for centralized processing and GPS context, LoRa for longer-range communication to a Base Unit, and an IoT/mobile interface for monitoring.

A distinctive element of the concept is the combination of an emergency flotation mechanism with an automatic rudder diversion system. The flotation subsystem is intended to support the wearer during a water emergency, while the rudder mechanism is intended to assist the vessel in moving toward an appropriate recovery direction under controlled conditions.

The estimated prototype cost is ₹58,000. Although the concept is technically promising at the prototype level, successful deployment would require extensive validation for marine environmental conditions, false alarms, actuator reliability, communications, power management, human factors and applicable safety/regulatory requirements.

Overall, the project establishes a foundation for further development into a connected marine safety platform. Future work should focus on prototype testing, miniaturization, marine-grade engineering, AI-assisted detection, improved vessel-control logic and collaboration with relevant marine-safety stakeholders.

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8. National and international marine safety guidance relevant to small vessels and personal flotation devices.
9. Relevant academic and technical literature on wearable sensors, IoT-based safety systems and marine communication.

28. APPENDICES

Appendix A – Suggested Smart Watch Data Packet

Field	Example
Device ID	W01
Event Type	SOS / MOTION / VITAL
Timestamp	YYYY-MM-DD HH:MM:SS
Heart Rate	BPM value
Motion Status	Normal / Alert
Battery	Percentage
Connection	BLE Connected

Appendix B – Suggested Emergency Packet

Field	Purpose
Boat ID	Identifies vessel
Wearer ID	Identifies affected person
Event Code	Defines emergency type
Latitude	Emergency location
Longitude	Emergency location
Timestamp	Event time
Battery Status	Device condition
Response Status	Pending / Activated / Closed

Appendix C – Prototype Demonstration Sequence

- Power on Smart Watch and Boat Hub.
- Register a wearable ID.
- Confirm the BLE connection.
- Display normal status.
- Trigger controlled SOS event.
- Verify Boat Hub receives an event.
- Verify that GPS context is attached.
- Verify LoRa alert reaches Base Unit.
- Demonstrate controlled flotation trigger using a safe test setup.
- Demonstrate rudder actuator on a bench/controlled model.
- Record response time and system logs.
- Reset system and repeat test.

Appendix D – Project Team

Team Member	Role / Contribution
Inbha Benisha J	Project development, documentation and presentation
Celano S	Technical development and prototype support
Fatima Rahana R	Project development, testing and documentation

Appendix E – Key Project Statement

Sense → Process → Detect → Communicate → Alert → Respond